

116 Advanced Statistics for Neuroscience

Exercise 01

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April 15, 2026

Bernoulli experiment

Use Matlab or Python or something similar to program a Bernoulli experiment with parameter $0 \leq p \leq 1$.

In other words, obtain a **random** outcome $x = 1$ with probability p or outcome $x = 0$ with probability $1 - p$.

Note that you will need to generate a random number from a suitable function.

Set of Bernoulli experiments

Reuse your Bernoulli function to perform a set of $n = 5$ identical but independent Bernoulli experiments, each with random outcome x_i , where $i = 1, \dots, n$.

The summed outcome

$$y = \sum_{i=1}^n x_i$$

will again be random but will cover a larger range, $y = 0, \dots, n$.
Note that there are $n + 1$ possible outcomes.

Repeated sets of Bernoulli experiments

Reuse your Set of Bernoulli function to repeat the experiment $N = 1000$ times and to collect the resulting outcomes y_k , where $k = 1, 2, \dots, N$ into a list (a Matlab vector or Numpy array).

Determine the proportion p_k of the $n + 1$ possible outcomes y , where $0 \leq y \leq n$, and plot

$$p_k \equiv \Pr(y = k - 1), \quad k = 1, 2, \dots, n + 1$$

and plot (visualize) the following functions:

- ▶ The probability function p_k (probability of each outcome y)
- ▶ The cumulative probability function $s_k = \sum_{k'=1}^k p_{k'}$, (probability of all outcomes up to and including k)

Check your results

Finally, check your results by plotting (additionally) the probability and cumulative probability of the **Binomial distribution** with parameters n and p , which you can obtain from suitable build-in functions.

In Python, you can use *binom.pmf* and *binom.cmf* after calling *from scipy.stats import binom*.

In Matlab, you can use built-in functions *binopdf* and *binocdf*.